

# Kun Xu

## Doctoral Researcher in Cyberspace Security

Ph.D. thesis defended | Degree conferral expected Oct. 2026

Trustworthy Generative AI and Multimedia Security

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Nanjing, Jiangsu 211106, China | Open to international postdoctoral opportunities



## Postdoctoral Research Profile

Doctoral researcher in Cyberspace Security at Nanjing University of Aeronautics and Astronautics (NUA), with the Ph.D. thesis successfully defended and formal degree conferral expected in October 2026. Research focuses on trustworthy generative AI, foundation-model evaluation, concept-level safety, uncertainty and calibration, privacy-aware synthesis, and multimedia forensics. Author of five published or accepted papers, with six additional manuscripts under review or submission. Recipient of a China Scholarship Council award supporting a one-year research visit in Italy. Seeking postdoctoral research opportunities in trustworthy AI, computer vision, multimedia security, and related areas, with an emphasis on reliable, secure, and responsible intelligent systems.

5

Published / accepted papers

6

Manuscripts under review / submission

12 mo.

CSC-funded research visit in Italy

## Research Expertise

- **Trustworthy generative AI and foundation-model evaluation:** concept-level safety, risk assessment, calibration, toxic and malicious content detection, and reliable governance of generative systems.
- **Diffusion-model uncertainty and representation:** uncertainty quantification, representation consistency, latent-space constraints, and generation-path guidance for personalized text-to-image models.
- **Privacy and multimedia forensics:** privacy-aware portrait release, deepfake detection and localization, adversarial defense, source-camera forensics, and medical-image integrity.

## International Research Experience

Sep. 2025 – Aug. 2026 **Visiting Ph.D. Researcher, University of Insubria** *Italy*  
Funded by the China Scholarship Council Excellence Talent Programme (International Training Programme for Outstanding Engineers). Supervised by Prof. Elena Ferrari (IEEE/ACM Fellow), with research collaboration involving Prof. Pierangela Samarati (IEEE/ACM Fellow) and Prof. Vincenzo Piuri (IEEE Fellow), University of Milan.

## Education

Sep. 2023 – Oct. 2026 **Ph.D. in Cyberspace Security** *Nanjing, China*  
Thesis successfully defended; formal degree conferral expected in October 2026. College of Computer Science and Technology, Nanjing University of Aeronautics and Astronautics. Advisor: Prof. Yushu Zhang.

Sep. 2020 – Jun. 2023 **Master's in Computer Science and Technology** *Huainan, China*  
School of Computer Science and Engineering, Anhui University of Science and Technology. Advisor: Prof. Gaoming Yang.

Sep. 2016 – Jun. 2020 **Bachelor's in Computer Science and Technology** *Huainan, China*  
School of Computer Science and Engineering, Anhui University of Science and Technology.

## Publications and Accepted Papers

- [1] **Kun Xu**, Wenying Wen\*, Shuren Qi, Tao Wang, Yushu Zhang, and Yuming Fang. “Detecting Malicious Concepts Without Image Generation in AI-Generated Content (AIGC).” *IEEE Transactions on Dependable and Secure Computing*, 23(4):8699–8716, 2026.
- [2] **Kun Xu**, Yushu Zhang\*, Tao Wang, Shuren Qi, Barbara Carminati, Elena Ferrari, and Yuming Fang. “Concept-Level Risk and Calibration for Governance in Diffusion Foundation Models.” In the *34th ACM International Conference on Multimedia (ACM MM)*, 2026. Accepted.

[3] **Kun Xu**, Shuren Qi, Yushu Zhang\*, Wenying Wen, and Hua Zhang. "A Facial Manipulation Adversarial Defense Approach for Image Post-Processing." *Computer Engineering & Science*, 47(12):2181–2194, 2025. In Chinese.

[4] **Kun Xu**, Gaoming Yang\*, Xianjin Fang, and Ji Zhang. "Facial Depth Forgery Detection Based on Image Gradient." *Multimedia Tools and Applications*, 82(19):29501–29525, 2023.

[5] Gaoming Yang, **Kun Xu\***, Xianjin Fang, and Ji Zhang. "Video Face Forgery Detection via Facial Motion-Assisted Capturing Dense Optical Flow Truncation." *The Visual Computer*, 39(11):5589–5608, 2023.

\* Corresponding author, as marked in the source publication list.

## Manuscripts Under Review or Submission

[1] **Kun Xu**, Yushu Zhang\*, Tao Wang, Shuren Qi, Zhongyun Hua, and Yuming Fang. "Understanding Concept-Driven Diffusion Model with Uncertainty." Manuscript under review at the *International Journal of Computer Vision*.

[2] **Kun Xu**, Yushu Zhang\*, Shuren Qi, Tao Wang, Barbara Carminati, Elena Ferrari, and Yuming Fang. "Identity Replacement Is Not Anonymization: Coupled-Condition Diffusion for Privacy-Aware Portrait Release." Manuscript under review at the *Network and Distributed System Security Symposium (NDSS)*.

[3] **Kun Xu**, Shuren Qi, Qing Tan, and Yushu Zhang\*. "Source-Camera-Aware Forgery Localization for IoT Visual Sensing from Photo-Response Non-Uniformity." Manuscript under review at the *Journal of Computer Science and Technology*.

[4] Wenying Wen, Zhen Chen, **Kun Xu**, Xiaojie Tao, and Yushu Zhang. "Detecting Subtle Lesion Tampering in Medical Images: A Deep Hashing Method with Multi-Scale Sparse Attention and Key-Weighted Mechanism." Manuscript under review at *IEEE Transactions on Information Forensics and Security*.

[5] Ziyue Yuan, Wenying Wen, **Kun Xu**, Tao Wang, and Yushu Zhang. "Having the Cake and Eating It Too: Differential-Response Hashing for Unified Integrity Authentication of Medical Images and Metadata." Manuscript under review at *IEEE Transactions on Image Processing*.

[6] Tao Wang, Yushu Zhang, Xiangli Xiao, **Kun Xu**, Lin Yuan, Wenying Wen, and Yuming Fang. "Rethinking Privacy Protection for Subject Faces in Photo Sharing: From Perturbation-Based to Synthesis-Based Methods." Manuscript under review at *IEEE Transactions on Information Forensics and Security*.

## Research Projects

|             |                                                                                                                                                     |                        |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
| 2021 – 2024 | <b>National Key R&amp;D Programme of China</b><br>"Blockchain-Governed Blockchain": cross-chain consensus algorithms and consistency proofs.        | Participant            |
| 2023 – 2026 | <b>Jiangxi Provincial Outstanding Young Scientists Fund Project</b><br>Key technologies for data security in multimedia Internet of Things systems. | Participant            |
| 2021 – 2023 | <b>High-End Foreign Experts Project, Ministry of Science and Technology</b><br>Machine-learning-based adversarial attacks and defenses.             | Participant            |
| 2022 – 2023 | <b>Innovation Project, Anhui University of Science and Technology</b><br>Detection and identification of deepfake face images and videos.           | Principal Investigator |

## Honors and Awards

|             |                                                                                                     |
|-------------|-----------------------------------------------------------------------------------------------------|
| 2025        | <b>Excellence Talent Programme National Scholarship</b><br>China Scholarship Council (CSC)          |
| 2024        | <b>National-Level Award, 3rd National College Student Data Analysis Popular Science Competition</b> |
| 2023 – 2025 | <b>Graduate Academic Scholarship</b><br>Nanjing University of Aeronautics and Astronautics          |
| 2021 – 2023 | <b>Graduate Academic Scholarship</b><br>Anhui University of Science and Technology                  |
| 2023        | <b>Outstanding Graduate of Anhui Province</b><br>Anhui Provincial Department of Education           |
| 2023        | <b>Outstanding Graduate</b><br>Anhui University of Science and Technology                           |

## Academic Service and Memberships

|                      |                                                                                                                                                                            |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Program committees   | ADFM 2026 @ CVPR; PEARC '26; SiMLA '26.                                                                                                                                    |
| Conference reviewing | ACM Multimedia 2026; ADFM 2026.                                                                                                                                            |
| Journal reviewing    | <i>Computer Vision and Image Understanding; Signal Processing; Journal of Computer Science and Technology; Multimedia Systems; Expert Systems; and Scientific Reports.</i> |
| Memberships          | ACM Emerging Interest Group on Trustworthy and Responsible Systems (EIGTRUST); China Society of Image and Graphics (CSIG).                                                 |

## Selected Talks and Research Exchanges

|           |                                                                                                                                                                                                              |
|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Nov. 2025 | <b>Research exchange at the University of Milan</b> <i>Milan, Italy</i><br>Academic discussion with Prof. Pierangela Samarati and Prof. Vincenzo Piuri on trustworthy AI and generative-model security.      |
| Oct. 2025 | <b>Minisymposium: Generative Concept Security in Trustworthy AIGC</b> <i>University of Insubria, Italy</i><br>Presentation connecting concept-level safety, uncertainty, and trustworthy generative systems. |

## Technical and Research Skills

|                      |                                                                                                                                                                                    |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Research             | Trustworthy AI, foundation-model evaluation, generative-model security, uncertainty quantification, deep learning, computer vision, multimedia forensics, and cyberspace security. |
| Programming          | Python and PyTorch; working knowledge of Java and C/C++.                                                                                                                           |
| ML and data tools    | NumPy, pandas, scikit-learn, Matplotlib, and OpenCV.                                                                                                                               |
| Systems and workflow | Linux, Git, Conda/Anaconda, reproducible experimentation, and scholarly writing.                                                                                                   |
| Mathematical methods | Probability, linear algebra, discrete mathematics, numerical analysis, and derivation of methods from the research literature.                                                     |
| Languages            | Mandarin Chinese and English.                                                                                                                                                      |

Publication and manuscript status is current as of September 2026.